

Network Security Labs and the Complete Group Linux Service Project

Adnan Khan

I completed practical network-security labs and contributed to our Group 22 Linux project. This walkthrough covers packet analysis, file-integrity checks and the full group service workflow, from the Fedora VM and user accounts to web/database services, FTP, time synchronization, mail-client setup, PGP and SELinux.

Tools and environment

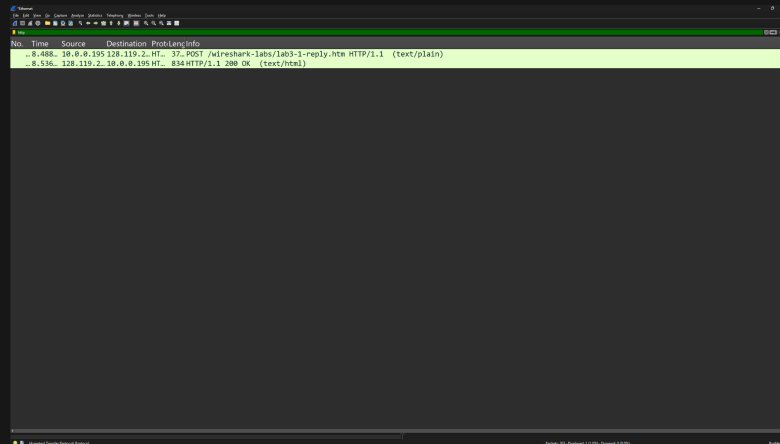
Wireshark; Windows command-line hash/checksum tools; PuTTY; TCP, HTTP, Telnet, SSH and TLS; VMware Workstation; Fedora; Apache, MariaDB, PHP, phpMyAdmin, Pure-FTPd, NTP, Evolution, Seahorse/GPG and SELinux.

Individual Assignments 1-4: Adnan Khan. Group 22: Abinayan Kokilan, Prathamdeep Multani, Adnan Khan and Tazwaar Alam. The group source does not allocate individual tasks. January-April 2026.

Follow a TCP upload from connection to data transfer

I identified client and server endpoints, followed SYN, SYN-ACK and ACK, and inspected the HTTP POST, acknowledgments and receive windows. An empty retransmission-filter view records no matches in that view. The assignment reported about 48 ms RTT and used 152,321 bytes over 0.236 seconds for its throughput calculation.

- The reported throughput inputs give approximately 645,428 bytes/s, or 5.16 Mbit/s. They are coursework measurements, not a new replay of the capture.



Client and server endpoints. Source portfolio p. 6.

68	8.308099	10.0.0.195	128.119.245.12	TCP	60	80	→	63086	[ACK]	Seq=1	Ack=81008	Win=112768	Len=0
69	8.308126	10.0.0.195	128.119.245.12	TCP	60	80	→	63086	[ACK]	Seq=1	Ack=82468	Win=115712	Len=0
71	8.341755	128.119.245.12	10.0.0.195	TCP	60	80	→	63086	[ACK]	Seq=1	Ack=83175	Win=118656	Len=0
72	8.341825	10.0.0.195	128.119.245.12	TCP	60	80	→	63086	[ACK]	Seq=1	Ack=84635	Win=121600	Len=0

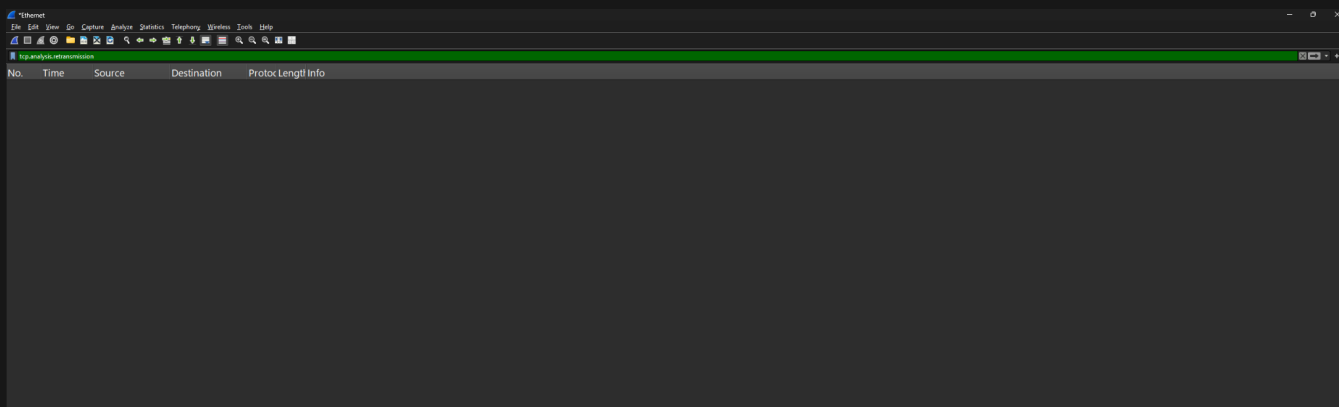
Tcp connection handshake. Source portfolio p. 8.

136	8.488234	10.0.0.195	128.119.245.12	HTTP	3783	POST	/wireshark-labs/lab3-1-reply.htm	HTTP/1.1	(text/plain)
-----	----------	------------	----------------	------	------	------	----------------------------------	----------	--------------

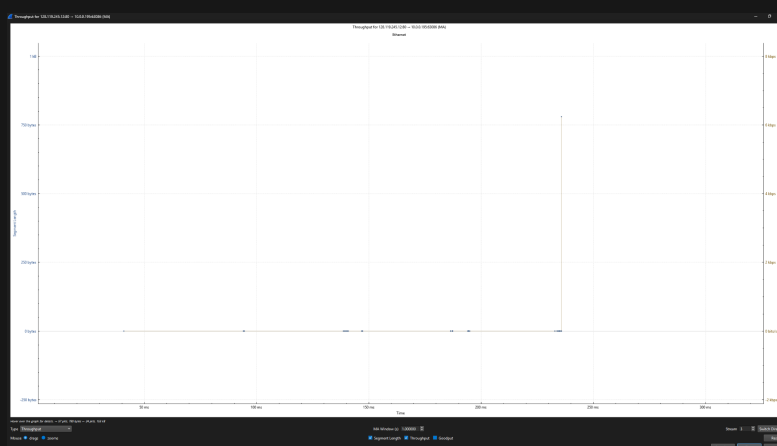
Http post and packet details. Source portfolio p. 9.

136	8.488234	10.0.0.195	128.119.245.12	HTTP	3783	POST	/wireshark-labs/lab3-1-reply.htm	HTTP/1.1	(text/plain)		
137	8.494889	128.119.245.12	10.0.0.195	TCP	60	80 → 63086	[ACK]	Seq=1	Ack=81008	Win=112768	Len=0
138	8.495188	128.119.245.12	10.0.0.195	TCP	60	80 → 63086	[ACK]	Seq=1	Ack=82468	Win=115712	Len=0
139	8.495188	128.119.245.12	10.0.0.195	TCP	60	80 → 63086	[ACK]	Seq=1	Ack=83175	Win=118656	Len=0
140	8.495188	128.119.245.12	10.0.0.195	TCP	60	80 → 63086	[ACK]	Seq=1	Ack=84635	Win=121600	Len=0
141	8.495392	128.119.245.12	10.0.0.195	TCP	60	80 → 63086	[ACK]	Seq=1	Ack=86095	Win=124544	Len=0

Acknowledgments and windows. Source portfolio p. 11.



Retransmission filter check. Source portfolio p. 13.



Transfer graph. Source portfolio p. 14.

Compare file integrity and HTTP authentication

I created ip.txt, calculated SHA-1/SHA-256 digests and a CRC checksum, changed the file and compared the values again. I also followed an HTTP 401 challenge and a later successful response. The exercise separates a content-change check from an authentication exchange.

- SHA-256 output changes when the file changes. CRC is an error-detection comparison; neither an untrusted hash nor reversible Base64 encoding establishes authenticity or confidentiality.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

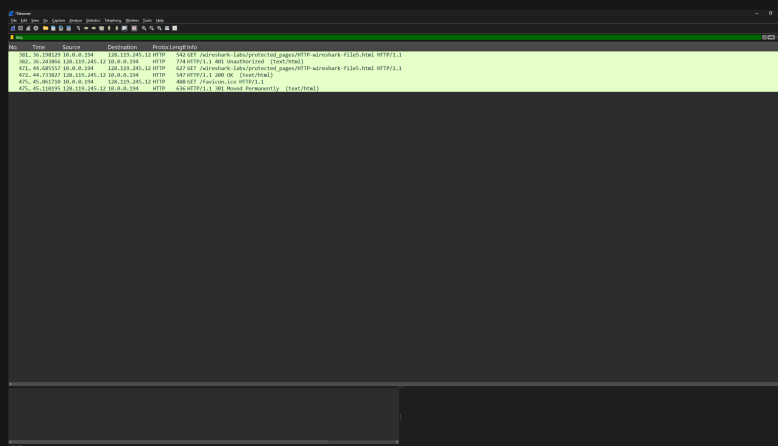
C:\Users\admin>shasum.exe ip.txt
                *ip.txt
                ip.txt
                *ip.txt

C:\Users\admin>ping ryerson.ca >> ip.txt

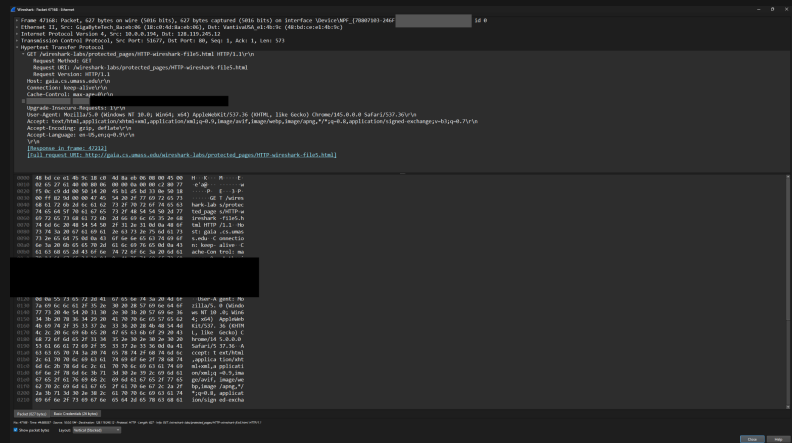
C:\Users\admin>shasum.exe ip.txt
                *ip.txt
                ip.txt
                *ip.txt

C:\Users\admin>
```

Detecting file changes. Source portfolio p. 20.



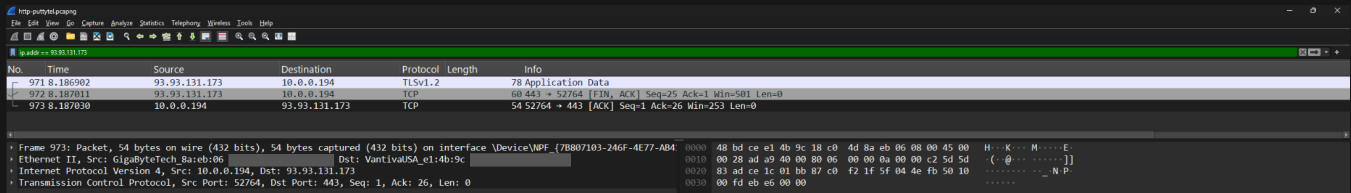
Http authentication exchange. Source portfolio p. 22.



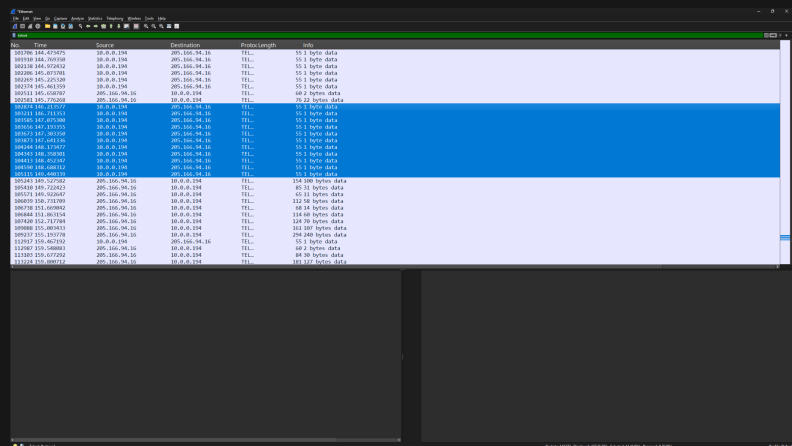
Http authentication exchange. Source portfolio p. 23.

Compare cleartext and encrypted sessions

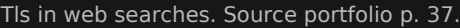
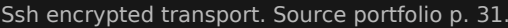
I compared Telnet and HTTP observations with SSH and TLS traffic. Cleartext protocol fields can expose content; encrypted packet views still reveal useful connection metadata. The selected SSH evidence identifies encrypted packets without presenting captured credentials.



Download connection capture. Source portfolio p. 26.

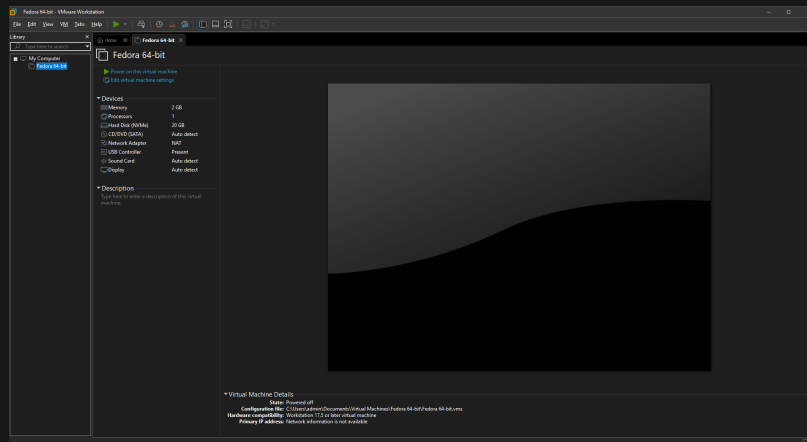


Telnet credential exposure. Source portfolio p. 28.

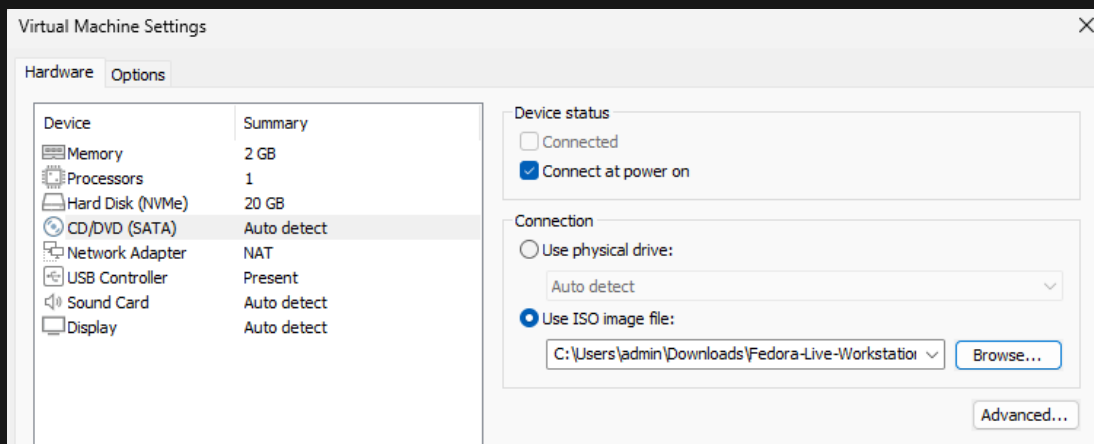


Build the group Linux environment and check networking

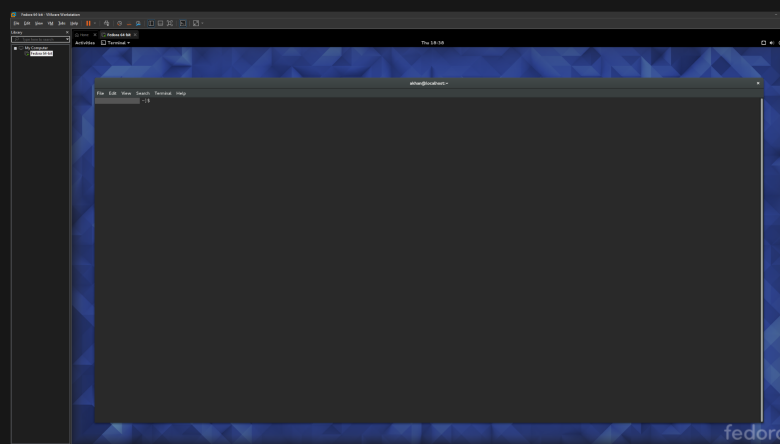
Our team documented VMware Workstation Pro 17 and a Fedora guest configured with 2 GB RAM, a 20 GB disk and NAT networking. We practised shell navigation, manual pages, hidden-file listings, interface inspection and DNS queries. Different later captures represent the coursework sequence rather than a verified single OS image.



Team evidence: Virtualization installation. Source portfolio p. 43.



Team evidence: Vm media and administrator account. Source portfolio p. 44.



Team evidence: Fedora installation and terminal. Source portfolio p. 47.

```

akhan]# ifconfig
eno16777984: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.140.129 netmask 255.255.255.0 broadcast 192.168.140.255
    inet6 fe80::20c:29ff:fe79:8ea1 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:79:8e:a1 txqueuelen 1000 (Ethernet)
    RX packets 349518 bytes  (496.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 57664 bytes 3380214 (3.2 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 
    loop txqueuelen 0 (Local Loopback)
    RX packets 4 bytes 340 (340.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 340 (340.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 192.168.124.1 netmask 255.255.255.0 broadcast 192.168.124.255
    ether txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

akhan]#

```

Team evidence: Linux interface inspection. Source portfolio p. 56.

```

akhan]# nslookup
> ryerson.ca
Server:      192.168.140.2
Address:     192.168.140.2#53

Non-authoritative answer:
Name:   ryerson.ca
Address: 141.117.126.20
> google.com
Server:      192.168.140.2
Address:     192.168.140.2#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.137.139
Name:   google.com
Address: 142.250.137.138
Name:   google.com
Address: 142.250.137.113
Name:   google.com
Address: 142.250.137.101
Name:   google.com
Address: 142.250.137.100
Name:   google.com
Address: 142.250.137.102
> exit
akhan]#

```

Team evidence: Linux dns queries. Source portfolio p. 58.

Create users, groups and inspect access

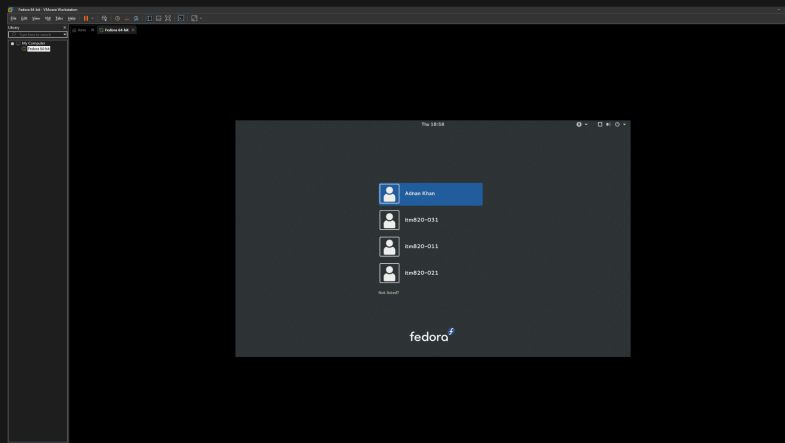
Our group created local accounts, used a project group and inspected membership and file ownership. A regular-user attempt to enter /root returned Permission denied, providing a direct access-boundary observation. The account and group work established the identity context for subsequent service exercises.

```

/]$ cd /root
bash: cd: /root: Permission denied
/]$ ls
bin    dev    home  lib64      media  opt    root  sbin  sys  usr
boot  etc    lib   lost+found mnt    proc  run   srv   tmp  var
/]$

```

Team evidence: Root directory and privilege context. Source portfolio p. 51.



Team evidence: Creating local users. Source portfolio p. 60.

```
itm820-011@localhost:/home/itm820-011
File Edit View Search Terminal Help
~]$ su
itm820-011]# groupadd project1
itm820-011]# lid -g project1
itm820-011]# usermod -a -G project1 itm820-011
itm820-011]# usermod -a -G project1 itm820-021
itm820-011]# usermod -a -G project1 itm820-031
itm820-011]# lid -g project1
itm820-011(uid=1001)
itm820-021(uid=1002)
itm820-031(uid=1003)
itm820-011]#
```

Team evidence: Groups and discretionary access. Source portfolio p. 61.

Build the Apache, MariaDB and PHP stack

Our team installed the web and database components, completed MariaDB initialization steps and installed PHP. The PHP information page records the PHP component responding through the web stack. Each component has its own setup evidence; a browser response is stronger than a package command alone.

```
File Edit View Search Terminal Help
~]$ su
akhan# yum install httpd
Yum command has been deprecated, redirecting to '/usr/bin/dnf install httpd'.
See 'man dnf' and 'man yum2dnf' for more information.
To transfer transaction metadata from yum to DNF, run:
'dnf install python-dnf-plugins-extras-migrate && dnf-2 migrate'

Last metadata expiration check performed 0:00:00 ago on Sat Apr  4 14:59:33 2026.
Package httpd-2.4.16-1.fc23.x86_64 is already installed, skipping.
Dependencies resolved.
=====
Package      Arch      Version      Repository      Size
=====
Skipping packages with conflicts:
(add '--best --allowerase' to command line to force their upgrade):
httpd        x86_64    2.4.23-4.fc23 updates        1.4 M
httpd-filesystem noarch    2.4.23-4.fc23 updates          26 k
httpd-tools  x86_64    2.4.23-4.fc23 updates          90 k
=====

Transaction Summary
=====
Nothing to do.
Complete!
akhan# systemctl start httpd
akhan# systemctl enable httpd
Created symlink from /etc/systemd/system/multi-user.target.wants/httpd.service to /usr/lib/systemd/system/httpd.service.
akhan#
```

Team evidence: Starting the web stack. Source portfolio p. 65.

```
akhan# yum install mariadb-server
Yum command has been deprecated, redirecting to '/usr/bin/dnf install mariadb-server'.
See 'man dnf' and 'man yum2dnf' for more information.
To transfer transaction metadata from yum to DNF, run:
'dnf install python-dnf-plugins-extras-migrate && dnf-2 migrate'

Last metadata expiration check performed 0:02:41 ago on Sat Apr  4 14:59:33 2026.
Dependencies resolved.
=====
Package      Arch      Version      Repository      Size
=====
Installing:
mariadb-connor x86_64    1:10.8.28-1.fc23 updates        5.9 M
mariadb-connor x86_64    1:10.8.28-1.fc23 updates          74 k
mariadb-errmsg x86_64    1:10.8.28-1.fc23 updates          25 k
mariadb-libs    x86_64    1:10.8.28-1.fc23 updates          159 k
mariadb-server x86_64    1:10.8.28-1.fc23 updates          18 M
perl-DBD-MariaDB x86_64    4.022-1.fc23 updates          132 k
perl-DBI         x86_64    1.020.0-1.fc23 Fedora           272 k
perl-Math-BigInt noarch    1.0007.340-1.fc23 Fedora          186 k
=====

Transaction Summary
=====
Install: 9 Packages
Total download size: 26 M
Installed size: 110 M
Is this ok [Y/n] y
Installing Packages
(1/9) mariadb-connor-1:10.8.28-1.fc23.x86_64.rpm
(2/9) mariadb-connor-1:10.8.28-1.fc23.x86_64.rpm
(3/9) mariadb-errmsg-1:10.8.28-1.fc23.x86_64.rpm
(4/9) mariadb-libs-1:10.8.28-1.fc23.x86_64.rpm
(5/9) perl-DBD-MariaDB-4.022-1.fc23.x86_64.rpm
(6/9) mariadb-server-1:10.8.28-1.fc23.x86_64.rpm
(7/9) perl-Math-BigInt-1.0007.340-1.fc23.noarch.rpm
(8/9) mariadb-libs-1:10.8.28-1.fc23.x86_64.rpm
(9/9) perl-DBI-1.020.0-1.fc23.x86_64.rpm
Total
-----
23 MiB | 20 MB/s | 00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Installing : mariadb-connor-1:10.8.28-1.fc23.x86_64
Installing : mariadb-connor-1:10.8.28-1.fc23.x86_64
Installing : mariadb-errmsg-1:10.8.28-1.fc23.x86_64
Installing : mariadb-libs-1:10.8.28-1.fc23.x86_64
Installing : perl-DBD-MariaDB-4.022-1.fc23.x86_64
Installing : mariadb-server-1:10.8.28-1.fc23.x86_64
Installing : perl-Math-BigInt-1.0007.340-1.fc23.noarch
Installing : mariadb-libs-1:10.8.28-1.fc23.x86_64
Installing : perl-DBI-1.020.0-1.fc23.x86_64
Verifying : mariadb-connor-1:10.8.28-1.fc23.x86_64
Verifying : mariadb-connor-1:10.8.28-1.fc23.x86_64
Verifying : mariadb-errmsg-1:10.8.28-1.fc23.x86_64
Verifying : mariadb-libs-1:10.8.28-1.fc23.x86_64
Verifying : perl-DBD-MariaDB-4.022-1.fc23.x86_64
Verifying : mariadb-server-1:10.8.28-1.fc23.x86_64
Verifying : perl-Math-BigInt-1.0007.340-1.fc23.noarch
Verifying : mariadb-libs-1:10.8.28-1.fc23.x86_64
Verifying : perl-DBI-1.020.0-1.fc23.x86_64
Total
-----
23 MiB | 20 MB/s | 00:02
Installed:
mariadb-connor.x86_64 1:10.8.28-1.fc23 mariadb-connor.x86_64 1:10.8.28-1.fc23 mariadb-errmsg.x86_64 1:10.8.28-1.fc23 mariadb-libs.x86_64 1:10.8.28-1.fc23 mariadb-server.x86_64 1:10.8.28-1.fc23
perl-DBD-MariaDB.x86_64 4.022-1.fc23 perl-DBI.x86_64 1.020.0-1.fc23 perl-Math-BigInt.noarch 1.0007.340-1.fc23
```

Team evidence: Database installation and setup. Source portfolio p. 66.

```
Disallow root login remotely? [Y/n] y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
akhan#
```

Team evidence: Database initialization checks. Source portfolio p. 69.

Team evidence: Php installation and validation. Source portfolio p. 70.

Team evidence: Php installation and validation. Source portfolio p. 71.

Add database administration and a protected web page

We installed phpMyAdmin and reached its login interface. The Apache exercise added directory authentication and an htpasswd account, restarted the service and tested the protected page. I distinguish the database login screen from the separate web-authentication exercise.



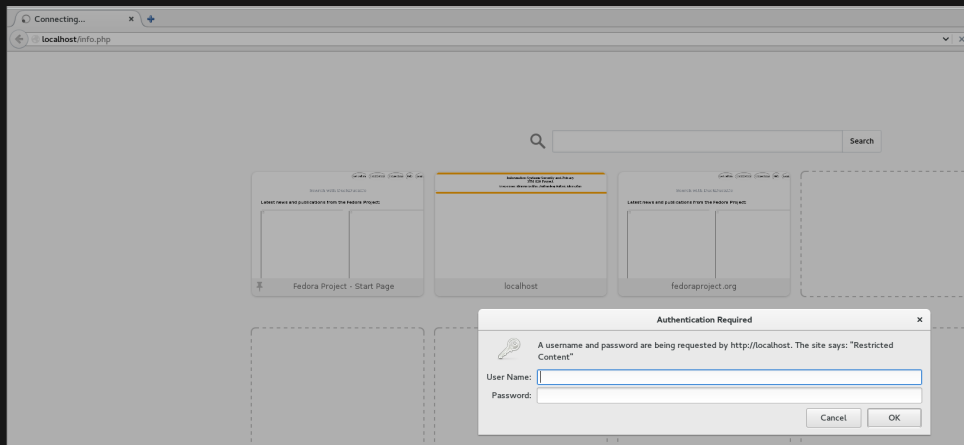
Team evidence: Database administration interface. Source portfolio p. 73.

```
AuthType Basic
AuthName "Restricted Content"
AuthUserFile /etc/httpd/.htpasswd
Require valid-user
```

Team evidence: Configuring web authentication. Source portfolio p. 77.

```
akhan]# htpasswd -c /etc/httpd/.htpasswd akhan
Adding password for user akhan
akhan]#
```

Team evidence: Testing the protected page. Source portfolio p. 78.



Team evidence: Testing the protected page. Source portfolio p. 80.

Information Systems Security and Privacy ITM 820 Project

Group names: Abinayan Kokilan, Prathamdeep Multani, Adnan Khan

Team evidence: Testing the protected page. Source portfolio p. 81.

Publish and inspect FTP test content

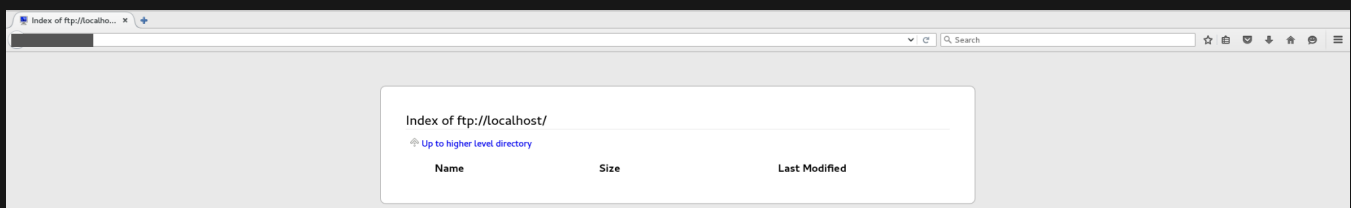
Our team installed and started Pure-FTPd, prepared content and browsed the local FTP directory. The exercise used anonymous access in the lab. The directory and file views are evidence of the coursework test, not a secure production file-transfer design.

```

ahel@ahel:~$ sudo yum install pure-ftpd
Last metadata expiration check performed 8:57:38 ago on Sat Apr 4 14:50:33 2020.
Dependencies resolved.
=====
Package Arch Version Repository Size
-----
Installing:
postgresql-libs x86_64 9.4.9-1.fc23 updates 240 k
pure-ftpd x86_64 1.0.42-3.fc23 updates 200 k
python x86_64 2.7.11-11.fc23 updates 6.9 k
python-libs x86_64 2.7.11-11.fc23 updates 1.9 k
python-pip noarch 7.1.0-1.fc23 Fedora 1.5 k
python-setuptools noarch 18.0.1-2.fc23 Fedora 425 k
Transaction Summary
-----
Install 6 Packages
Total download size: 8.4 M
Installed size: 34 M
Downloading Packages:
(1/6): python-2.7.11-11.fc23.x86_64.rpm 460 kB/s | 95 kB 00:00
(2/6): postgresql-libs-9.4.9-1.fc23.x86_64.rpm 978 kB/s | 240 kB 00:00
(3/6): pure-ftpd-1.0.42-3.fc23.x86_64.rpm 1.3 MB/s | 200 kB 00:00
(4/6): python-setuptools-18.0.1-2.fc23.noarch.rpm 4.0 MB/s | 425 kB 00:00
(5/6): python-pip-7.1.0-1.fc23.noarch.rpm 12 MB/s | 1.5 kB 00:00
(6/6): python-libs-2.7.11-11.fc23.x86_64.rpm 22 MB/s | 5.8 MB 00:00
Total: 6.4 MB/s | 8.4 MB 00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Installing : python-libs-2.7.11-11.fc23.x86_64 1/6
Installing : python-pip-7.1.0-1.fc23.noarch 2/6
Installing : python-setuptools-18.0.1-2.fc23.noarch 3/6
Installing : python-2.7.11-11.fc23.x86_64 4/6
Installing : postgresql-libs-9.4.9-1.fc23.x86_64 5/6
Installing : pure-ftpd-1.0.42-3.fc23.x86_64 6/6
Verifying : pure-ftpd-1.0.42-3.fc23.x86_64 1/6
Verifying : postgresql-libs-9.4.9-1.fc23.x86_64 2/6
Verifying : python-2.7.11-11.fc23.x86_64 3/6
Verifying : python-libs-2.7.11-11.fc23.x86_64 4/6
Verifying : python-pip-7.1.0-1.fc23.noarch 5/6
Verifying : python-setuptools-18.0.1-2.fc23.noarch 6/6
Installed:
  postgresql-libs.x86_64 9.4.9-1.fc23      pure-ftpd.x86_64 1.0.42-3.fc23      python.x86_64 2.7.11-11.fc23      python-libs.x86_64 2.7.11-11.fc23      python-pip.noarch 7.1.0-1.fc23      python-setuptools.noarch 18.0.1-2.fc23
Complete!
[root@localhost ahel]#

```

Team evidence: Ftp service and test content. Source portfolio p. 82.



Team evidence: Ftp service and test content. Source portfolio p. 84.

Index of ftp://localhost/

[Up to higher level directory](#)

Name	Size	Last Modified
 pub		04/04/26 04:01:00 PM

Team evidence: Browsing the ftp directory. Source portfolio p. 87.

Index of ftp://localhost/pub/Group No 01 - ITM 820/

[Up to higher level directory](#)

Name	Size	Last Modified
 HelloWorld.txt	1 KB	04/04/26 04:03:00 PM

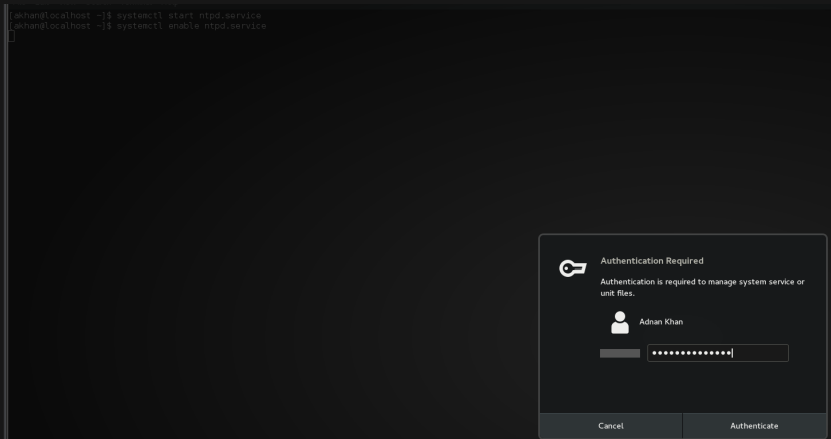
Team evidence: Browsing the ftp directory. Source portfolio p. 89.

Configure time synchronization

We installed NTP, configured Fedora pool entries with `iburst`, started and enabled `ntpd`, and queried its peers. The peer output gives a recorded service check after configuration. Time-service setup belongs alongside the other operational dependencies of the lab.

```
server 0.fedora.pool.ntp.org iburst
server 1.fedora.pool.ntp.org iburst
server 2.fedora.pool.ntp.org iburst
server 3.fedora.pool.ntp.org iburst
```

Team evidence: Configuring time synchronization. Source portfolio p. 91.



Team evidence: Configuring time synchronization. Source portfolio p. 92.

```

~]$ systemctl start ntpd.service
~]$ systemctl enable ntpd.service
Created symlink from /etc/systemd/system/multi-user.target.wants/ntpd.service to /usr/lib/systemd/system/ntpd.service.
~]$ ntpq -p
      remote           refid      st t when poll reach  delay  offset  jitter
=====
*zero.gotrroot.ca 192.80.172.163  2 u  47   64    1  65.460 -1.074  1.955
167.160.187.12    167.160.187.179 4 u  46   64    1  29.814 -9.695  2.269
archer.fsck.ca   206.108.0.133   2 u  45   64    1  24.348  1.190  1.440
time3.mbix.ca    .PPS.           1 u  44   64    1  36.646 -2.569  2.505
64.ip-54-39-23.  51.79.53.181    3 u  43   64    1  22.792 -1.252  3.403
clock.yxeix.ca   198.169.133.150 2 u  42   64    1  47.139 -0.844  2.459
mirror3.rafal.c  85.133.38.92    2 u  41   64    1  22.397 -0.191  5.197
backoffice-1.in  206.108.0.133   2 u  40   64    1  51.664  1.885  3.709
~]$

```

Team evidence: Configuring time synchronization. Source portfolio p. 93.

Prepare the mail client and PGP identities

Our group checked the mail-related service, installed Evolution and configured account settings using POP on port 995 and SMTP on port 465. We also used Seahorse for PGP identities, key lists and keyserver lookup. These are configuration and key-management exercises; the source does not establish successful message delivery.

```

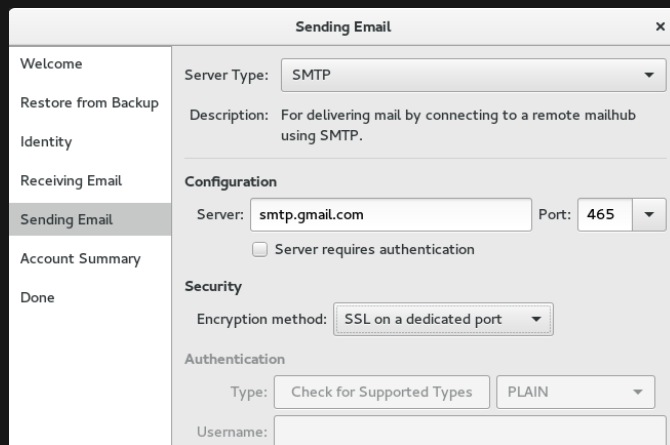
sbin]$ su
Password:
sbin# yum install evolution
Yum command has been deprecated, redirecting to '/usr/bin/dnf install evolution'.
See 'man dnf' and 'man yum2dnf' for more information.
To transfer transaction metadata from yum to DNF, run:
'dnf install python-dnf-plugins-extras-migrate && dnf-2 migrate'

Last metadata expiration check performed 1:20:38 ago on Sat Apr  4 14:59:33 2026.
Package evolution-3.18.1-1.fc23.x86_64 is already installed, skipping.
Dependencies resolved.

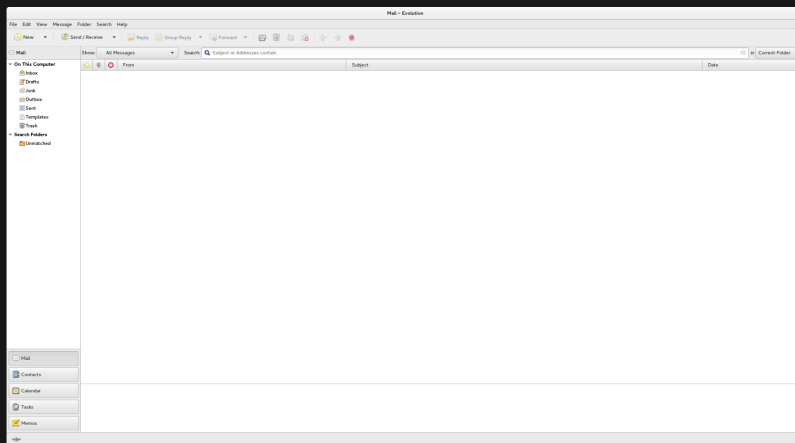
=====
Package                                Arch                                Version                                Repository                                Size
=====
Skipping packages with conflicts:
(add '--allowrasing' to command line to force their upgrade):
evolution                               x86_64                               3.18.5-2-1.fc23                        updates                                    9.6 M
evolution-help                          noarch                               3.18.5-2-1.fc23                        updates                                    2.1 M
Transaction Summary
-----
Nothing to do.
Complete!
sbin#

```

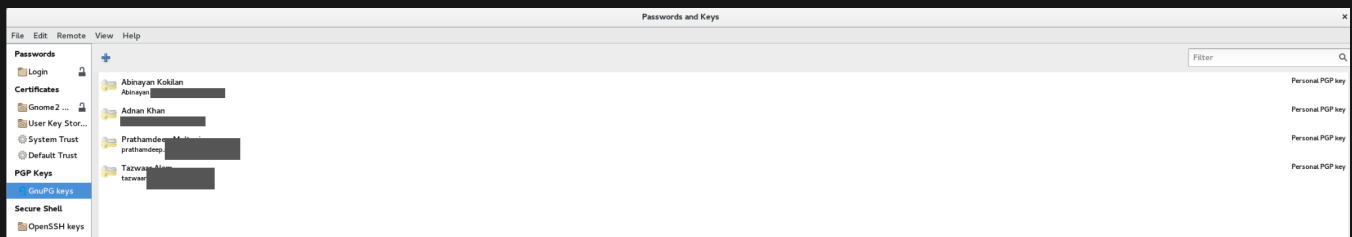
Team evidence: Mail client preparation. Source portfolio p. 95.



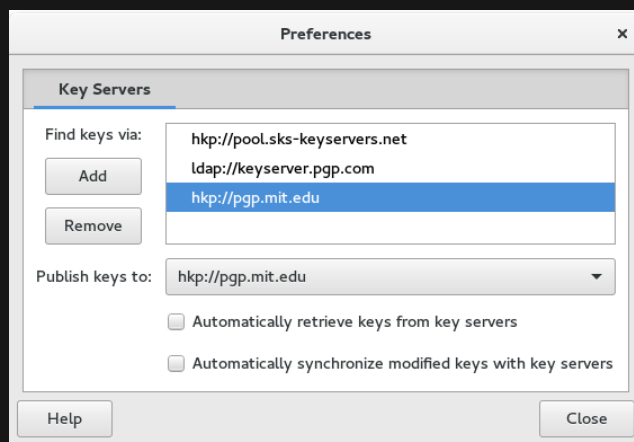
Team evidence: Pop and smtp account settings. Source portfolio p. 99.



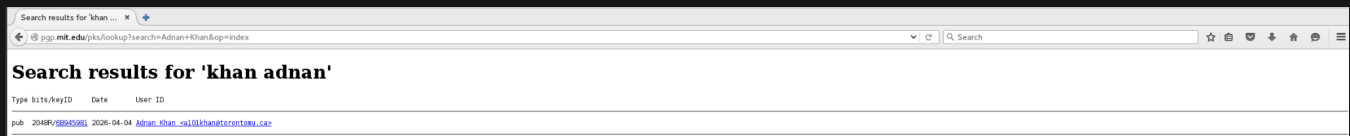
Team evidence: Mail account summary. Source portfolio p. 101.



Team evidence: Group key identities. Source portfolio p. 107.



Team evidence: Keyserver configuration and lookup. Source portfolio p. 108.



Team evidence: Keyserver configuration and lookup. Source portfolio p. 110.

Inspect SELinux policy and login mappings

We installed policy tools, inspected sestatus and mapped the lab login to user_u with semanage. The resulting table records range s0. I describe the visible configuration as a targeted-policy/login-mapping exercise rather than an MLS enforcement result.

```
[7/8] python3-policycoreutils-0:3.9-7.fc43.noarch 100% | 0.0 B/s | 0.0 B | 00m00s
>>> Already downloaded
[8/8] policycoreutils-python-utils-0:3.9-7.fc43.noarch 100% | 0.0 B/s | 0.0 B | 00m00s
>>> Already downloaded
-----
[8/8] Total 100% | 0.0 B/s | 0.0 B | 00m00s
Running transaction
[ 1/15] Verify package files 100% | 85.0 B/s | 8.0 B | 00m00s
[ 2/15] Prepare transaction 100% | 44.0 B/s | 13.0 B | 00m00s
[ 3/15] Upgrading policycoreutils-0:3.9-7.fc43.x86_64 100% | 6.1 MiB/s | 713.8 KiB | 00m00s
[ 4/15] Upgrading selinux-policy-0:43.4-2.fc43.noarch 100% | 4.2 KiB/s | 33.6 KiB | 00m08s
[ 5/15] Upgrading selinux-policy-targeted-0:43.4-2.fc43.noarch 100% | 50.3 MiB/s | 14.8 MiB | 00m00s
[ 6/15] Installing policycoreutils-newrole-0:3.9-7.fc43.x86_64 100% | 1.3 MiB/s | 34.0 KiB | 00m00s
[ 7/15] Upgrading python3-policycoreutils-0:3.9-7.fc43.noarch 100% | 62.0 MiB/s | 5.9 MiB | 00m00s
[ 8/15] Installing mcstrans-0:3.9-2.fc43.x86_64 100% | 1.9 MiB/s | 318.5 KiB | 00m00s
[ 9/15] Installing selinux-policy-mls-0:43.4-2.fc43.noarch 100% | 58.3 MiB/s | 18.2 MiB | 00m00s
[10/15] Upgrading policycoreutils-python-utils-0:3.9-7.fc43.noarch 100% | 1.6 MiB/s | 98.0 KiB | 00m00s
[11/15] Removing selinux-policy-0:42.12-1.fc43.noarch 100% | 428.0 B/s | 12.0 B | 00m00s
[12/15] Removing selinux-policy-targeted-0:42.12-1.fc43.noarch 100% | 37.5 KiB/s | 1.7 KiB | 00m00s
[13/15] Removing policycoreutils-python-utils-0:3.9-5.fc43.noarch 100% | 1.5 KiB/s | 24.0 B | 00m00s
[14/15] Removing python3-policycoreutils-0:3.9-5.fc43.noarch 100% | 7.7 KiB/s | 386.0 B | 00m00s
[15/15] Removing policycoreutils-0:3.9-5.fc43.x86_64 100% | 6.0 B/s | 188.0 B | 00m26s
Complete!
```

Team evidence: Installing selinux policy tools. Source portfolio p. 113.

```
/home/liveuser# sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:       /etc/selinux
Loaded policy name:            targeted
Current mode:                  enforcing
Mode from config file:         enforcing
Policy MLS status:             enabled
Policy deny_unknown status:    allowed
Memory protection checking:    actual (secure)
Max kernel policy version:     35
/home/liveuser#
```

Team evidence: Checking selinux policy and mode. Source portfolio p. 114.

		semanage login -a -s user_u group22	
		semanage login -l	
Login Name	SELinux User	MLS/MCS Range	Service
__default__	unconfined_u	s0-s0:c0.c1023	*
group22	user_u	s0	*
root	unconfined_u	s0-s0:c0.c1023	*
	/home/liveuser#		

Team evidence: Mapping a linux login to selinux. Source portfolio p. 118.

Bring the security and service checks together

The individual labs examine traffic and integrity. The team project adds accounts and services around those concepts. I used the evidence to distinguish setup, a reachable interface and a verified operation, rather than treating all three as the same result.

- Troubleshooting: a denied /root operation required checking the current identity.
- Packet analysis: connection flags, HTTP status and encrypted payloads answer different questions.
- Service validation: mail delivery, decryption round trips and policy-enforcement tests would require additional evidence.

What I learned

- Follow the evidence from transport setup to the application response.
- Check identity, ownership and service state together when investigating access.
- A service installation and a successful end-to-end transaction are different milestones.

Source: the full ITM820 portfolio supplied for this project. Evidence is from the original coursework; this edition adds no new lab test results. Repeated captures and earlier drafts are consolidated. Revised design summaries are labelled, and sensitive identifiers are masked.